

Q1 - 24 June - Shift 1

The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is :

- (A) 9 (B) 10
(C) 11 (D) 12

Space for your notes:

Q2 - 24 June - Shift 1

$f(x) = 4 \log_e(x - 1) - 2x^2 + 4x + 5$, $x > 1$, which one of the following is NOT correct ?

- (A) f is increasing in $(1, 2)$ and decreasing in $(2, \infty)$
(B) $f(x) = -1$ has exactly two solutions
(C) $f'(e) - f''(2) < 0$
(D) $f(x) = 0$ has a root in the interval $(e, e + 1)$

Space for your notes:

Q3 - 24 June - Shift 1

the tangent at the point (x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve :

- (A) $x^2 + \frac{y^2}{81} = 2$ (B) $\frac{y^2}{9} - x^2 = 8$
(C) $y = 4x^2 + 5$ (D) $\frac{x}{3} - y^2 = 2$

Space for your notes:

Q4 - 24 June - Shift 1

Questions

MathonGo

The sum of absolute maximum and absolute minimum values of the function

$f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ in the interval

$[0, 1]$ is :

(A) $3 + \frac{\sin(1) \cos^2(\frac{1}{2})}{2}$ (B) $3 + \frac{1}{2}(1 + 2\cos(1)) \sin(1)$

(C) $5 + \frac{1}{2}(\sin(1) + \sin(2))$ (D) $2 + \sin\left(\frac{1}{2}\right) \cos\left(\frac{1}{2}\right)$

Space for your notes:

Q5 - 24 June - Shift 2

The number of distinct real roots of the equation

$x^7 - 7x - 2 = 0$ is

(A) 5 (B) 7 (C) 1 (D) 3

Space for your notes:

Q6 - 24 June - Shift 2

Let λ^* be the largest value of λ for which the

function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is

increasing for all $x \in \mathbb{R}$. Then $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ is

equal to :

(A) 36 (B) 48

(C) 64 (D) 72

Space for your notes:

Q7 - 25 June - Shift 1

Questions

MathonGo

Let $f(x)$ be a polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value

of $\lim_{x \rightarrow 1} \frac{f(x)}{x-1}$

(A) -15 (B) -60

(C) 60 (D) 15

Space for your notes:

Q8 - 25 June - Shift 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and

$g(x) = \frac{1 - 2e^{2x}}{e^x}$. Then, for which of the

following range of α , the inequality

$f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds?

(A) $(2, 3)$ (B) $(-2, -1)$

(C) $(1, 2)$ (D) $(-1, 1)$

Space for your notes:

Q9 - 25 June - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

Water is being filled at the rate of $1 \text{ cm}^3 / \text{sec}$ in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is

- (A) 5 (B) $\frac{\sqrt{21}}{5}$
(C) $\frac{\sqrt{26}}{5}$ (D) $\frac{\sqrt{26}}{10}$

Space for your notes:

Q10 - 25 June - Shift 2

If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12(t + \sin t \cos t)$, $y = 12(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to

- (A) $6(3 + 2\sqrt{2})$ (B) $3(7 + 4\sqrt{3})$
(C) 27 (D) 48

Space for your notes:

Q11 - 25 June - Shift 2

Let $f(x) = |(x-1)(x^2 - 2x - 3)| + x - 3$, $x \in \mathbb{R}$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval $(0, 4)$, then $m + M$ is equal to _____

Space for your notes:

#MathBoleTohMathonGo

Q12 - 26 June - Shift 1

The sum of the absolute minimum and the absolute maximum values of the function $f(x) = |3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is :

- (A) $\frac{\sqrt{17}+3}{2}$ (B) $\frac{\sqrt{17}+5}{2}$
 (C) 5 (D) $\frac{9-\sqrt{17}}{2}$

Space for your notes:

Q13 - 26 June - Shift 1

Let S be the set of all the natural numbers, for which the line $\frac{x}{a} + \frac{y}{b} = 2$ is a tangent to the curve

$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ at the point (a, b) , $ab \neq 0$. Then:

- (A) $S = \phi$ (B) $n(S) = 1$
 (C) $S = \{2k : k \in \mathbb{N}\}$ (D) $S = \mathbb{N}$

Space for your notes:

Q14 - 26 June - Shift 1

Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If $[a, b]$ is the range of the function then $4a - b$ is equal to:

- (A) 11 (B) $11 - \pi$ (C) $11 + \pi$ (D) $15 - \pi$

Space for your notes:

Q15 - 26 June - Shift 2

Consider a cuboid of sides $2x$, $4x$ and $5x$ and a closed hemisphere of radius r . If the sum of their surface areas is a constant k , then the ratio $x : r$, for which the sum of their volumes is maximum, is :

- (A) $2 : 5$ (B) $19 : 45$ (C) $3 : 8$ (D) $19 : 15$

Space for your notes:

Questions

MathonGo

Q16 - 26 June - Shift 2

If $y = y(x)$ is the solution of the differential equation $x \frac{dy}{dx} + 2y = xe^x, y(1) = 0$, then the local maximum value of the function $z(x) = x^2 y(x) - e^x, x \in \mathbb{R}$ is :

- (A) $1 - e$ (B) 0 (C) $\frac{1}{2}$ (D) $\frac{4}{e} - e$

Space for your notes:

Q17 - 27 June - Shift 1

The number of distinct real roots of $x^4 - 4x + 1 = 0$ is :

- (A) 4 (B) 2
(C) 1 (D) 0

Space for your notes:

Q18 - 27 June - Shift 1

The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k^2$ is equal to :

- (A) 5 (B) 8
(C) 10 (D) 12

Space for your notes:

Q19 - 27 June - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

If m and n respectively are the number of local maximum and local minimum points of the function $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$, then the ordered pair (m, n) is equal to

- (A) (3, 2) (B) (2, 3)
(C) (2, 2) (D) (3, 4)

*Space for your notes:***Q20 - 28 June - Shift 1**

The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____.

- (A) 0 (B) 1
(C) 3 (D) 5

*Space for your notes:***Q21 - 28 June - Shift 1**

Let ℓ be a line which is normal to the curve $y = 2x^2 + x + 2$ at a point P on the curve. If the point $Q(6, 4)$ lies on the line ℓ and O is origin, then the area of the triangle OPQ is equal to _____.

*Space for your notes:***Q22 - 28 June - Shift 2**

Questions

MathonGo

Let the slope of the tangent to a curve $y = f(x)$ at (x, y) be given by $2 \tan x (\cos x - y)$. If the curve passes through the point $(\pi/4, 0)$, then the value

of $\int_0^{\pi/2} y dx$ is equal to

- (A) $(2 - \sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (B) $2 - \frac{\pi}{\sqrt{2}}$
(C) $(2 + \sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (D) $2 + \frac{\pi}{\sqrt{2}}$

Q23 - 29 June - Shift 1

A wire of length 22 m is to be cut into two pieces.

One of the pieces is to be made into a square and the other into an equilateral triangle. Then, the length of the side of the equilateral triangle, so that the combined area of the square and the equilateral triangle is minimum, is :

- (A) $\frac{22}{9+4\sqrt{3}}$ (B) $\frac{66}{9+4\sqrt{3}}$
(C) $\frac{22}{4+9\sqrt{3}}$ (D) $\frac{66}{4+9\sqrt{3}}$

Q24 - 29 June - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$f(x) = (x-3)^{n_1} (x-5)^{n_2}$, $n_1, n_2 \in \mathbb{N}$. The, which of

the following is NOT true?

(A) For $n_1 = 3$, $n_2 = 4$, there exists $\alpha \in (3,5)$ where

f attains local maxima.

(B) For $n_1 = 4$, $n_2 = 3$, there exists $\alpha \in (3,5)$ where f attains local minima.

(C) For $n_1 = 3$, $n_2 = 5$, there exists $\alpha \in (3,5)$ where f attains local maxima.

(D) For $n_1 = 4$, $n_2 = 6$, there exists $\alpha \in (3,5)$ where f attains local maxima.

Space for your notes:

Q25 - 29 June - Shift 2

Let f and g be twice differentiable even functions

on $(-2, 2)$ such that $f\left(\frac{1}{4}\right) = 0, f\left(\frac{1}{2}\right) = 0, f(1) = 1$

and $g\left(\frac{3}{4}\right) = 0, g(1) = 2$ Then, the minimum number

of solutions of $f(x) g''(x) + f'(x) g'(x) = 0$ in $(-2,2)$ is equal to__.

Space for your notes:

Answer Key

Q1 (A)

Q2 (C)

Q3 (D)

Q4 (B)

Q5 (D)

Q6 (D)

Q7 (A)

Q8 (A)

Q9 (C)

Q10 (C)

Q11 (3)

Q12 (A)

Q13 (D)

Q14 (B)

Q15 (B)

Q16 (D)

Q17 (B)

Q18 (C)

Q19 (B)

Q20 (B)

Q21 (13)

Q22 (B)

Q23 (B)

Q24 (C)

Q25 (4)

#MathBoleTohMathonGo

Q1 (A)Let r be the radius of spherical balloon S = Surface area

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = 8\pi r \times \frac{dr}{dt} = k \text{ (constant)}$$

$$4\pi r^2 = kt + C \text{ (C is constant of integration)}$$

$$\text{For } t = 0, r = 3 \Rightarrow 36\pi = C$$

$$\text{For } t = 5, r = 7 \Rightarrow K = 32\pi$$

$$4\pi r^2 = 32\pi t + 36\pi$$

$$r^2 = 8t + 9$$

$$\text{for } t = 9$$

$$r^2 = 81$$

$$r = 9$$

Q2 (C)

$$f(x) = 4 \log_e(x-1) - 2x^2 + 4x + 5, x > 1$$

$$f'(x) = \frac{4}{x-1} - 4(x-1)$$

$$\text{For } 1 < x < 2 \Rightarrow f'(x) > 0$$

$$\text{For } x > 2 \Rightarrow f'(x) < 0 \text{ (option 1 is correct)}$$

$$f(x) = -1 \text{ has two solution (option 2 is correct)}$$

$$f(e) > 0$$

$$f(e+1) < 0$$

$$f(e).f(e+1) < 0 \text{ (option 4 is correct)}$$

$$f'(e) - f''(2) = \frac{4}{e-1} - 4(e-1) + 8 > 0$$

$$\text{(option C is incorrect)}$$

Q3 (D)

The tangent at (x_1, y_1) to the curve

$$y = x^3 + 3x^2 + 5$$

$$y - y_1 = (3x_1^2 + 6x_1)(x - x_1) \text{ passing through origin}$$

$$-y_1 = (3x_1^3 + 6x_1^2)(-x_1)$$

$$y_1 = (3x_1^3 + 6x_1^2) \text{ -----(1)}$$

And (x_1, y_1) lies on the curve

$$y = x^3 + 3x^2 + 5$$

$$y_1 = x_1^3 + 3x_1^2 + 5 \text{ -----(2)}$$

From equation (1) and (2)

$$2y_1 = 3x_1^2 + \frac{15}{2}$$

Hence the equation of curve $y = \frac{3}{2}x^2 + \frac{15}{2}$

This curve does not intersect $\frac{x}{3} - y^2 = 2$

Q4 (B)

$$f(x) = |2x^2 + 3x - 2| + \sin x \cos x$$

$$f(x) = |(2x - 1)(x + 2)| + \sin x \cos x$$

$$f'(x) = \begin{cases} 4x + 3 + \frac{\cos 2x}{4}, & \frac{1}{2} < x < 1 \\ -(4x + 3) + \frac{\cos 2x}{4}, & 0 \leq x < \frac{1}{2} \end{cases}$$

$$\text{For } 0 \leq x < \frac{1}{2} \Rightarrow f'(x) < 0$$

$$\text{For } \frac{1}{2} < x \leq 1 \Rightarrow f'(x) > 0$$

$$f(x) \text{ local minima at } x = \frac{1}{2} \text{ and}$$

$$\text{local maxima at } x = 1$$

$$f\left(\frac{1}{2}\right) + f(1) = 3 + \frac{1}{2}(1 + 2\cos 1) \sin 1$$

Q5 (D)

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Hints and Solutions

MathonGo

$$x^7 - 7x - 2 = 0$$

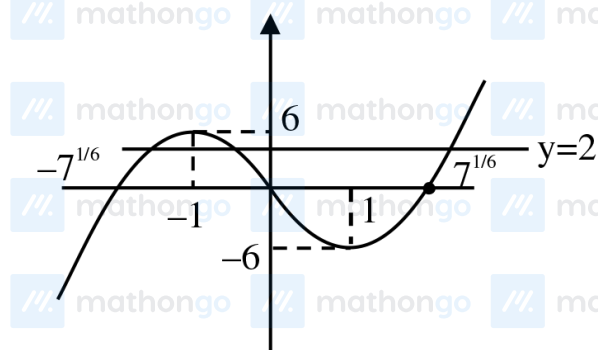
$$x^7 - 7x = 2$$

$$f(x) = x^7 - 7x \text{ (odd) \& } y = 2$$

$$f(x) = x(x^2 - 7^{1/3})(x^4 + x^2 \cdot 7^{1/3} + 7^{2/3})$$

$$f'(x) = 7(x^6 - 1) = 7(x^2 - 1)(x^4 + x^2 + 1)$$

$$f'(x) = 0 \Rightarrow x = \pm 1$$



$f(x) = 2$ has 3 real distinct solution.

Q6 (D)

#MathBoleTohMathonGo

$$f_{\lambda}(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$$

$$f'_{\lambda}(x) = 12\lambda x^2 - 72\lambda x + 36$$

$$f'_{\lambda}(x) = 12(\lambda x^2 - 6\lambda x + 3) \geq 0$$

$$\therefore \lambda > 0 \text{ \& } D \leq 0$$

$$36\lambda^2 - 4 \times \lambda \times 3 \leq 0$$

$$9\lambda^2 - 3\lambda \leq 0$$

$$3\lambda(3\lambda - 1) \leq 0$$

$$\lambda \in \left[0, \frac{1}{3}\right]$$

$$\therefore \lambda_{\text{largest}} = \frac{1}{3}$$

$$f(x) = \frac{4}{3}x^3 - 12x^2 + 36x + 48$$

$$\therefore f(1) + f(1) = 72$$

Q7 (A)

$$\lim_{x \rightarrow 1} \frac{f(x)}{x-1} = f'(1) \text{ (and } f(1) = 0)$$

$$f(x) + f'(x) + f''(x) = x^5 + 64$$

$$f'(x) + f''(x) + f'''(x) = 5x^4$$

$$f''(x) + f'''(x) + f^{iv}(x) = 20x^3$$

$$f'''(x) + f^{iv}(x) + f^v(x) = 60x^2$$

$$\therefore f^v(x) - f''(x) = 60x^2 - 20x^3$$

$$\Rightarrow 120 - f''(1) = 40 \Rightarrow f''(1) = 80$$

$$\text{Also } f(1) + f'(1) + f''(1) = 65 \Rightarrow f'(1) = -15. \text{ Ans.}$$

#MathBoleTohMathonGo

Q8 (A)

$$f(x) = \log_e(x^2 + 1) - e^{-x} + 1$$

$$\Rightarrow f'(x) = \frac{2x}{x^2 + 1} + e^{-x} > 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow f$ is strictly increasing

$$g(x) = \frac{1 - 2e^{2x}}{e^x} = e^{-x} - 2e^x$$

$$\Rightarrow g'(x) = -(2e^x + e^{-x}) < 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow g$ is decreasing

$$\text{Now } f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$$

$$\Rightarrow g\left(\frac{(\alpha-1)^2}{3}\right) > g\left(\alpha - \frac{5}{3}\right)$$

$$\Rightarrow \frac{(\alpha-1)^2}{3} < \alpha - \frac{5}{3}$$

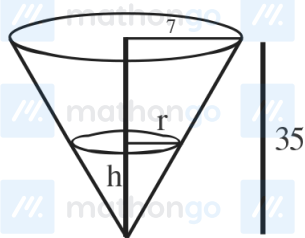
$$\Rightarrow \alpha^2 - 5\alpha + 6 < 0$$

$$\Rightarrow (\alpha - 2)(\alpha - 3) < 0$$

$$\Rightarrow \alpha \in (2, 3)$$

Q9 (C)

From figure $\frac{r}{h} = \frac{7}{35} \Rightarrow h = 5r$



$$\text{Given } \frac{dV}{dt} = 1 \Rightarrow \frac{d}{dt} \left(\frac{\pi r^2 h}{3} \right) = 1$$

$$\Rightarrow \frac{d}{dt} \left(\frac{5\pi}{3} r^3 \right) = 1 \Rightarrow r^2 \frac{dr}{dt} = \frac{1}{5\pi}$$

Let wet conical surface area = S

$$= \pi r \ell = \pi r \sqrt{h^2 + r^2}$$

$$= \sqrt{26} \pi r^2 \Rightarrow \frac{dS}{dt} = 2\sqrt{26} \pi r \frac{dr}{dt}$$

$$\text{When } h = 10 \text{ then } r = 2 \Rightarrow \frac{dS}{dt} = \frac{2\sqrt{26}}{10}$$

Q10 (C)

$$\frac{dy}{dx} = \frac{2(1 + \sin t) \times \cos t}{1 + \cos 2t}$$

$$\Rightarrow \frac{2(1 + \sin t) \cos t}{2 \cos^2 t} = \sqrt{3}$$

$$\Rightarrow t = \frac{\pi}{6}, y_0 = 27$$

Q11 (3)

$$f(x) = \begin{cases} (x^2 - 1)(x - 3) + (x - 3), & x \in (0, 1] \cup [3, 4) \\ -(x^2 - 1)(x - 3) + (x - 3), & x \in [1, 3] \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 3x^2 - 6x, & x \in (0, 1) \cup (3, 4) \\ -3x^2 + 6x + 2, & x \in (1, 3) \end{cases}$$

$f(x)$ is non-derivable at $x = 1$ and $x = 3$

$$\text{also } f'(x) = 0 \text{ at } x = 1 + \sqrt{\frac{5}{3}} \Rightarrow m + M = 3$$

Q12 (A)

$$f(x) = \begin{cases} x^2 - 4x - 2, & \forall x \in \left(-1, \frac{3 - \sqrt{17}}{2}\right) \\ -x^2 + 2x + 2, & \forall x \in \left(\frac{3 - \sqrt{17}}{2}, 2\right) \end{cases}$$

$$f'(x) \text{ when } x \in \left(-1, \frac{3 - \sqrt{17}}{2}\right)$$

$$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$$

$$f'(x) = 2(x - 2) \Rightarrow f'(x) \text{ is always } \downarrow$$

$$f(2) = 2$$

$$f(-1) = 3$$

$$f\left(\frac{3 - \sqrt{17}}{2}\right) = \frac{\sqrt{17} - 3}{2}$$

$$f'(x) \text{ when } x \in \left(\frac{3 - \sqrt{17}}{2}, 2\right)$$

$$f'(x) = -2x + 2$$

$$f'(x) = -2(x - 1)$$

$$f'(x) = 0 \text{ when } x = 1$$

$$f(1) = 3$$

$$\text{absolute minimum value} = \frac{\sqrt{17} - 3}{2}$$

$$\text{absolute maximum value} = 3$$

$$\text{Sum} = \frac{\sqrt{17} - 3}{2} + 3 = \frac{\sqrt{17} + 3}{2}$$

Q13 (D)

$$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$$

Slope of tangent at (a, b)

$$n \cdot \left(\frac{x}{a}\right)^{n-1} \cdot \frac{1}{a} + n \left(\frac{y}{b}\right)^{n-1} \cdot \frac{1}{b} \frac{dy}{dx} = 0$$

$$\left. \frac{dy}{dx} \right|_{(a,b)} = -\frac{b}{a}$$

 $\therefore$ Equation of tangent

$$y - b = -\frac{b}{a}(x - a)$$

$$\frac{x}{a} + \frac{y}{b} = 2 \quad \forall n \in \mathbb{N}$$

Q14 (B)

$$f'(x) = \frac{-2}{\sqrt{1-x^2}} - \frac{4}{1+x^2} - 6x - 2$$

$$= -2 \left[\frac{1}{\sqrt{1-x^2}} + \frac{2}{1+x^2} + 3x + 1 \right]$$

 $f'(x) < 0 \Rightarrow f(x)$ is a dec. function

$$f(1) = \pi + 5$$

$$f(-1) = 5\pi + 5$$

$$\text{Range : } [a, b] = [\pi + 5, 5\pi + 5]$$

$$a = \pi + 5, b = 5\pi + 5 \Rightarrow 4a - b = 11 - \pi.$$

Q15 (B)

Hints and Solutions

MathonGo

$$\text{Surface area} = 76x^2 + 3\pi r^2 = \text{constant (K)}$$

$$V = 40x^3 + \frac{2}{3}\pi r^3$$

$$[76x^2 + 3\pi r^2 = K]$$

$$r^2 = \frac{K - 76x^2}{3\pi}$$

$$r = \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}}$$

$$V = 40x^3 + \frac{2}{3}\pi \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{3}{2}}$$

$$\frac{dV}{dx} = 120x^2 + \frac{2}{3}\pi \cdot \frac{3}{2} \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}} \cdot \left(\frac{-76(2x)}{3\pi} \right)$$

Put

$$\frac{dV}{dx} = 0 \Rightarrow 120x^2 + \frac{2}{3}\pi \cdot \frac{3}{2} \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}} \cdot \left(\frac{-76(2x)}{3\pi} \right) = 0$$

$$\Rightarrow 120x^2 = \frac{152x}{3} \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}}$$

$$\Rightarrow \frac{45}{19}x^2 = x \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}}; x \neq 0$$

$$\Rightarrow \frac{45}{19}x = \left(\frac{K - 76x^2}{3\pi} \right)^{\frac{1}{2}} \Rightarrow \left(\frac{45}{19} \right)^2 x^2 = \frac{K - 76x^2}{3\pi}$$

$$\Rightarrow \left(\frac{45}{19} \right)^2 x^2 = r^2 \Rightarrow \frac{x^2}{r^2} = \left(\frac{19}{45} \right)^2$$

$$\Rightarrow \frac{x}{r} = \frac{19}{45}$$

Q16 (D)

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$$x \frac{dy}{dx} + 2y = xe^x$$

$$\frac{dy}{dx} + \frac{2y}{x} = e^x$$

$$\text{I.F.} = x^2$$

$$y \cdot x^2 = \int x^2 e^x dx$$

$$= \int e^x (x^2 + 2x - 2x - 2 + 2) dx$$

$$yx^2 = e^x (x^2 - 2x + 2) + c$$

$$y(1) = 0$$

$$0 = e(1 + 0) + c$$

$$c = -e$$

$$z(x) = x^2 y(x) - e^x$$

$$= e^x (x^2 - 2x + 2) - e - e^x$$

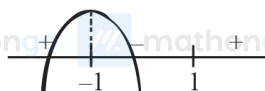
$$= e^x (x - 1)^2 - e$$

$$\frac{dz}{dx} = e^x \cdot 2(x - 1) + e^x (x - 1)^2 = 0$$

$$x^x (x - 1) (2 + x - 1) = 0$$

$$e^x (x - 1) (x + 1) = 0$$

$$x = -1, 1$$



$x = -1$ local maxima. Then maximum value is

$$z(-1) = \frac{4}{e} - e$$

Q17 (B)

Hints and Solutions

MathonGo

Let $f(x) = x^4 - 4x + 1$

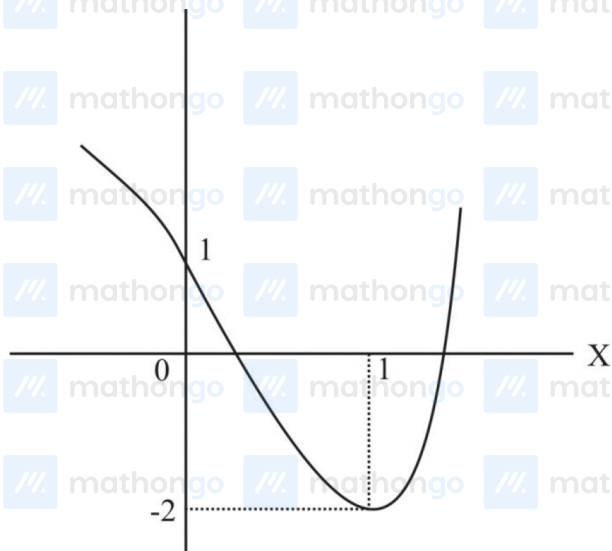
$$f'(x) = 4x^3 - 4$$

$$f'(x) = 0 \Rightarrow x = 1$$

$x = 1$ is point of minima.

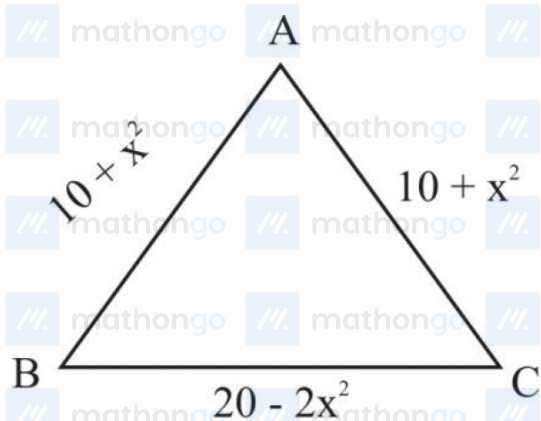
$$f(1) = -2$$

$$f(0) = 1$$



Hence 2 solutions.

Q18 (C)



$$a = 20 - 2x^2, b = 10 + x^2, c = 10 + x^2$$

$$= \frac{a+b+c}{2}$$

$$= 20$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{20(2x^2)(10-x^2)(10-x^2)}$$

$$= 2\sqrt{10} \sqrt{x^2(10-x^2)^2}$$

$$= 2\sqrt{10} |x(10-x^2)|$$

$$= 2\sqrt{10} |10x - x^3|$$

$$S = 10x - x^3$$

$$\frac{ds}{dx} = 10 - 3x^2$$

$$\frac{ds}{dx} = 0 \Rightarrow x^2 = \frac{10}{3}$$

$$3x^2 = 10$$

Q19 (B)

Hints and Solutions

MathonGo

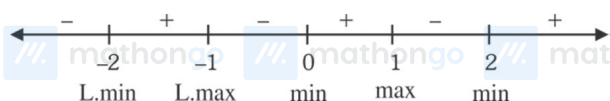
$$m = L \cdot \max$$

$$N = L \cdot \min$$

$$f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$$

$$f'(x) = \frac{(x^4 - 5x^2 + 4)2x}{2 + e^{x^2}} = \frac{2x(x^2 - 1)(x^2 - 4)}{2 + e^{x^2}}$$

$$= \frac{2x(x-1)(x+1)(x-2)(x+2)}{2 + e^{x^2}}$$



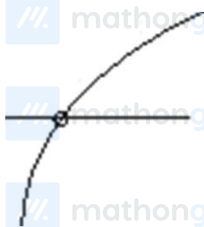
$$\text{so, } m = 2 \quad \text{and} \quad n = 3$$

Q20 (B)

$$f(x) = x^7 + 5x^3 + 3x + 1$$

$$f'(x) = 7x^6 + 15x^2 + 3 > 0$$

$\therefore f(x)$ is strictly increasing function



$$x \rightarrow -\infty, y \rightarrow -\infty$$

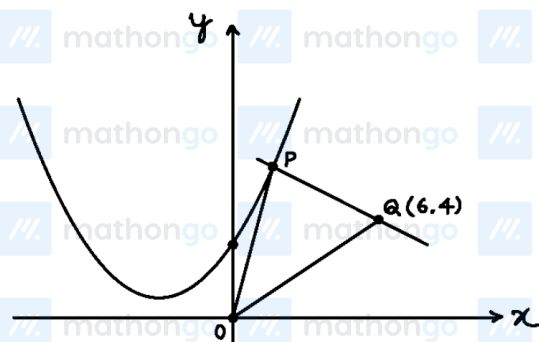
$$x \rightarrow \infty, y \rightarrow \infty$$

$\therefore$ no. of real solution = 1

Q21 (13)

#MathBoleTohMathonGo

$$y = 2x^2 + x + 2$$



$$\frac{dy}{dx} = 4x + 1$$

Let P be (h, k), then normal at P is

$$y - k = -\frac{1}{4h + 1}(x - h)$$

This passes through Q (6, 4)

$$\therefore 4 - k = -\frac{1}{4h + 1}(6 - h)$$

$$\Rightarrow (4h + 1)(4 - k) + 6 - h = 0$$

$$\text{Also } k = 2h^2 + h + 2$$

$$\therefore (4h + 1)(4 - 2h^2 - h - 2) + 6 + h = 0$$

$$\Rightarrow 4h^3 - 3h^2 + 3h - 8 = 0$$

$$\Rightarrow h = 1, k = 5$$

$$\text{Now area of } \triangle OPQ \text{ will be} = \frac{1}{2} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 5 \\ 1 & 6 & 4 \end{vmatrix} = 13$$

Q22 (B)

$$\frac{dy}{dx} = 2 \tan x \cos x - 2 \tan x \cdot y$$

$$\frac{dy}{dx} + (2 \tan x) y = 2 \sin x$$

$$\text{Integrating factor} = e^{\int 2 \tan x dx} = \frac{1}{\cos^2 x}$$

$$y \left(\frac{1}{\cos^2 x} \right) = \int \frac{2 \sin x}{\cos^2 x} dx$$

$$y \sec^2 x = \frac{2}{\cos x} + C$$

$$y = 2 \cos x + C \cos^2 x$$

$$\text{Passes through } \left(\frac{\pi}{4}, 0 \right)$$

$$0 = \sqrt{2} + \frac{C}{2} \Rightarrow C = -2\sqrt{2}$$

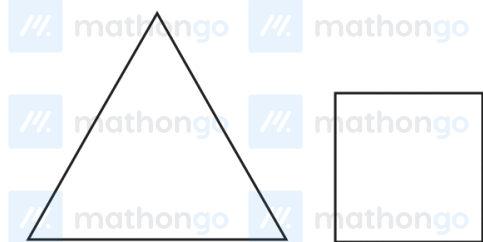
$$f(x) = 2 \cos x - 2\sqrt{2} \cos^2 x : \text{Required curve}$$

$$\int_0^{\pi/2} y dx = 2 \int_0^{\pi/2} \cos x dx - 2\sqrt{2} \int_0^{\pi/2} \cos^2 x dx$$

$$= [2 \sin x]_0^{\pi/2} - 2\sqrt{2} \left[\frac{x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi/2}$$

$$= 2 - \frac{\pi}{\sqrt{2}}$$

Q23 (B)



$$3a = x$$

$$4b = 22 - x$$

$$a = 2/13$$

$$A_T = \frac{\sqrt{3}}{4} a^2 + b^2$$

$$= \frac{\sqrt{3}}{4} x^2 / 9 + \frac{(22-x)^2}{16}$$

$$\frac{dA}{dx} = 0 \Rightarrow x \left(\frac{\sqrt{3}}{2 \times 9} + \frac{1}{8} \right) - \frac{22}{8} = 0$$

$$\Rightarrow x \left(\frac{4\sqrt{3} + 9}{36} \right) = \frac{11}{2}$$

$$a = x/3$$

$$a = \left(\frac{11/2}{4\sqrt{3} + 9} \right) \left(\frac{1}{3} \right) = \frac{66}{4\sqrt{3} + 9}$$

Q24 (C)

$$f'(x) = (x-3)^{n_1-1} (x-5)^{n_2-1} (n_1 + n_2) \left(x - \frac{5n_1 + 3n_2}{n_1 + n_2} \right)$$

Option (3) is incorrect since

$$\text{for } n_1 = 3, n_2 = 5$$

$$f(x) = 8(x-3)^2(x-5)^4 \left(x - \frac{30}{8} \right)$$

$$\text{minima at } x = \frac{30}{8}$$

Q25 (4)

#MathBoleTohMathonGo

Let $h(x) = f(x)g'(x) \rightarrow 5$ roots

$\therefore f(x)$ is even $\Rightarrow$

$$f\left(\frac{1}{4}\right) = f\left(\frac{1}{2}\right) = f\left(-\frac{1}{2}\right) = f\left(\frac{1}{4}\right) = 0$$

$$g(x) \text{ is even} \Rightarrow g\left(\frac{3}{4}\right) = g\left(-\frac{3}{4}\right) = 0$$

$g'(x) = 0$ has minimum one root

$h'(x)$ has at last 4 roots